Assignment 1 — Univariate and Multivariate Statistics

FinTech 545 — Quantitative Risk Management

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1 Conventions

Every number quoted here is printed by the script named at the head of the section, and every figure is written by the same script. `README.md` gives the commands.

Two conventions are used throughout, because packages disagree about both:

- **Variance** uses the $n-1$ denominator. **Skewness and excess kurtosis** use the bias corrected estimators, and kurtosis is always reported in *excess* form, so a normal sample returns 0 rather than 3. `scipy.stats` defaults to the biased estimators, so `bias=False` is passed explicitly everywhere.
- **AICc** uses $k = p + d$, following Week 02: p counts the regression parameters including the intercept, d counts the parameters of the error distribution. A one regressor model with a normal error has $k = 3$; with a Student's t error it has $k = 4$. `statsmodels` reports AIC only, so the correction term $\frac{2k^2+2k}{n-k-1}$ is added by hand.

2 Reading the Shape of a Sample

Code: `problem1.py`.

2.1 Predict

The first four moments of `problem1.csv`, $n = 1000$:

Table 1: First four moments of the sample

Moment	Value
Mean	0.001104
Variance	9.624822×10^{-5}
Standard deviation	0.009811
Skewness	-0.6698
Excess kurtosis	$+2.3576$

The biased estimators give -0.6688 and $+2.3398$; at $n = 1000$ the correction is immaterial, but the convention still has to be stated because a package reporting raw rather than excess kurtosis would have printed 5.36 and changed the reading entirely.

The pair is feasible: every distribution satisfies kurtosis $\geq$ skewness$^2 + 1$, and here $5.3576 \geq 1.4487$.

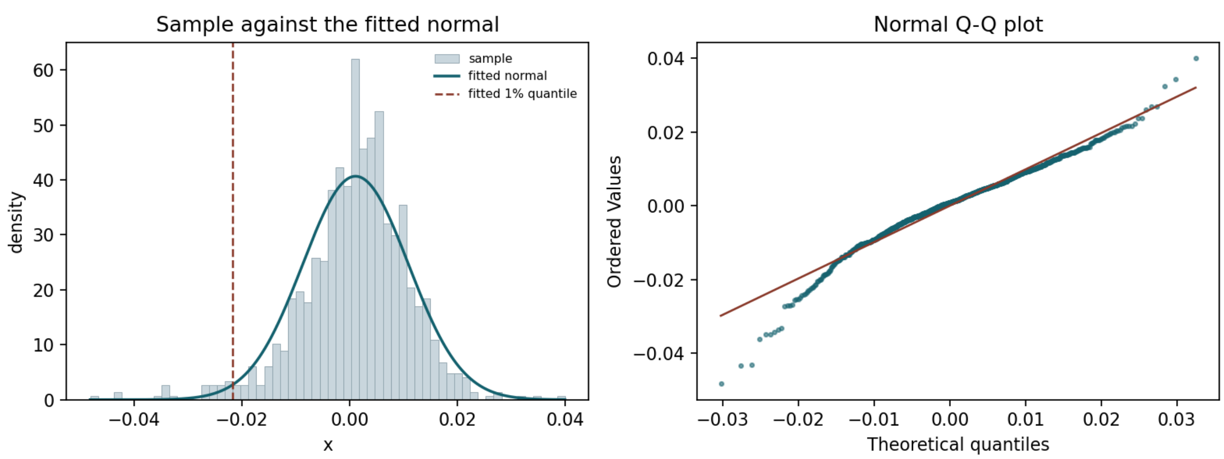

Figure 1: Left: the sample against the normal fitted by matching its mean and variance, with that normal's 1% quantile marked. Right: the normal Q-Q plot. Both panels show the same thing — the body is described well and both tails leave the line, the left one much further than the right.

(a) Which Week 1 distributions could plausibly have produced this sample?

The sample is **negatively skewed with fat tails**. Taking the four families from Week 01 in turn, and naming the moment that does the rejecting:

- **Normal — rejected, by the third moment and independently by the fourth.** The normal has skewness 0 and excess kurtosis 0 at every μ and σ. The sample has -0.67 and $+2.36$. Either one alone is fatal; both fire. This is not something fitting can repair, because neither moment is a free parameter of the family.
- **Lognormal — rejected, by the third moment.** The lognormal has skewness $(e^{\sigma^2} + 2)\sqrt{e^{\sigma^2} - 1}$, which is *strictly positive* for every $\sigma > 0$. A negative sample skewness cannot be produced by any member of the family. A second reason, independent of any moment: the lognormal is supported on $(0, \infty)$ and 406 of the 1000 observations are negative.
- **Student's t — rejected, by the third moment.** The t is symmetric, so its skewness is 0 whenever it exists ($\nu > 3$). It handles the fourth moment comfortably — $6/(\nu - 4) = 2.36$ gives $\nu \approx 6.5$ — but it leaves the skew entirely unexplained. This is the family that gets the tails right and the asymmetry wrong.
- **Normal inverse Gaussian — survives.** Its skewness is $3\beta/(\alpha\sqrt{\delta\gamma})$, which takes the sign of β, so $\beta < 0$ delivers negative skew; its excess kurtosis $\frac{3}{\delta\gamma}\left(1 + \frac{4\beta^2}{\alpha^2}\right)$ is strictly positive. It is the only one of the four that can carry both signs at once.

So of the four families, only the NIG is admissible. By the shape table in Week 01 a skewed t or a generalized hyperbolic would also survive. Two moments narrow the field; they do not pick a winner.

2.2 Fit and reconcile

Matching the mean and the variance gives $N(0.001104,\ 0.009811^2)$. Its 1% quantile is -0.021719.

(b) How many observations fall below the fitted normal's 1% quantile?

26. Under the fitted model the count is $\text{Binomial}(1000, 0.01)$, so the expected number is **10** with a standard deviation of 3.15. The observed count is 5.09 standard deviations above expectation. This is not sampling noise.

(c) What would the fitted normal get wrong, where, and in which direction?

Both tails, in opposite directions. Counting exceedances at four levels:

Table 2: Exceedances of the fitted normal, both tails

α	Lower observed	Lower expected	Upper observed	Upper expected
0.005	18	5.0	5	5.0
0.010	26	10.0	6	10.0
0.025	40	25.0	15	25.0
0.050	50	50.0	34	50.0

Three things follow, and they are exactly the two rejected moments made visible.

The left tail is understated, and the error grows as you go further out. At the 5% level the count is exact, 50 against 50. At 2.5% it is 1.6× too many, at 1% 2.6×, at 0.5% 3.6×. The body of the distribution is fine; the model fails only in the tail, and it fails worse the further into the tail you look. That is the signature of the excess kurtosis in Table 1.

The right tail is overstated. 6 observations above the 99% quantile against 10 expected, 34 above the 95% against 50. The normal predicts *more* large gains than occurred.

The asymmetry between those two is the skewness. A symmetric fat tailed model would have missed on both sides in the same direction. Here the model understates the left and overstates the right, which is precisely what a symmetric family fitted to a left skewed sample must do: forced to be symmetric, it splits the difference and is wrong in opposite directions on the two sides.

For risk management this is the dangerous orientation of the error. If this series is a return series, the fitted normal underestimates the frequency of large losses by a factor of roughly three at the levels a capital calculation cares about, while simultaneously flattering the upside.

3 A Regression Whose Errors Are Not Normal

Code: `problem2.py`.

3.1 Predict

(a) What does the scatter show, and what error distribution do I expect?

The left panel of Figure 2 shows a strong, clean positive linear relationship — a slope near 3 is readable by eye — with two features that matter. The vertical spread does not widen or narrow as x moves, so there is no fanning. But a handful of points sit far off the line in both directions, much further than the bulk of the scatter would suggest.

That combination — constant spread, occasional large deviations, roughly symmetric — is a **fat tailed but not heteroskedastic error**. I expect a Student's t to describe it and a normal not to.

The residual moments from the OLS fit confirm the reading before any distributional model is fitted: skewness -0.3624, excess kurtosis $+3.6168$. Excess kurtosis of $+3.6$ is a lot; $6/(\nu-4) = 3.6$ implies $\nu \approx 5.7$.

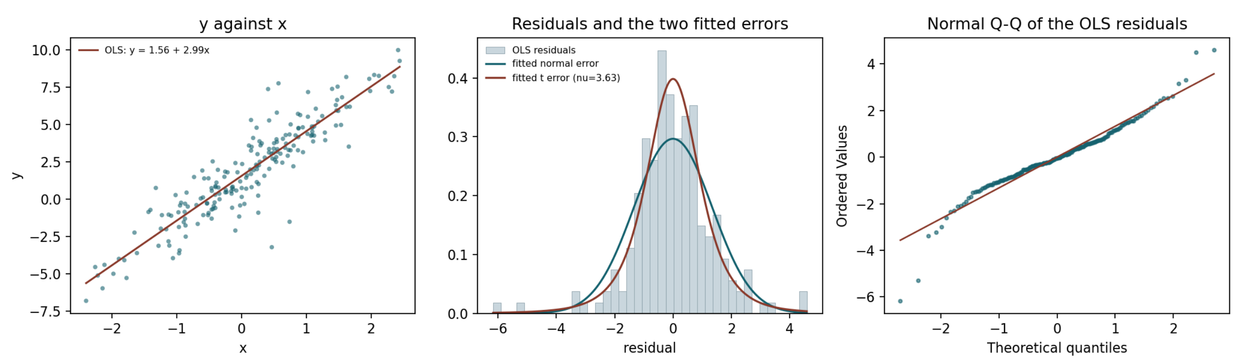

Figure 2: Left: `y` against `x` with the OLS line. Centre: the OLS residuals against the two fitted error densities. Right: the normal Q-Q plot of the residuals, whose S-shape is the fat tails.

(b) Which of the seven OLS assumptions does this violate, and what will the violation do?

Assumption 7, that the error term is normally distributed. The other candidates can be ruled out with the standard diagnostics:

Table 3: Which assumption actually fails

Assumption	Test	Result
4, uncorrelated errors	Durbin–Watson	2.0159, no serial correlation
5, constant variance	Breusch–Pagan	$p = 0.7798$, no heteroskedasticity
7, normal errors	Jarque–Bera	106.23, $p = 8.6 \times 10^{-24}$

Assumption 1 holds by construction (the model is linear in the parameters), assumption 6 cannot fail with a single regressor, and assumptions 2 and 3 follow in sample from including an intercept and fitting by least squares. Only normality rejects, and it rejects overwhelmingly.

My predictions for what that does:

To the estimated slope: almost nothing. The unbiasedness of OLS requires assumptions 1 through 3 and nothing about the shape of the error. Normality is only needed to make the OLS solution *also* the maximum likelihood estimator and to justify the inference built on top of it. So I expect the three slope estimates to be close, and I expect the difference between them to be small relative to one OLS standard error.

To the standard error of that slope: real damage. The formula $s\sqrt{\text{diag}((X'X)^{-1})}$ is derived under normality. Two separate things go wrong when the error is fat tailed. First, s^2 is an average of squared residuals, so the few very large residuals inflate it, and the reported standard error is larger than an estimator that handled the tails properly would need. Second, and more importantly, the reported number is no longer the standard deviation of the sampling distribution of $\hat{\beta}$, so the t statistics and the confidence intervals built from it are not trustworthy at their stated levels.

3.2 Fit

Table 4: Three models fitted to `problem2.csv`, $n = 200$

Model	α	β	Error parameters	$\ln L$	k	AICc
OLS	1.5607	2.9908	$s = 1.3514$	-343.0110	3	692.1445
MLE, normal	1.5607	2.9908	$\sigma = 1.3446$	-343.0110	3	692.1445
MLE, t	1.5480	3.0192	$\sigma = 0.9355$, $\nu = 3.6322$	-328.5612	4	**665.3275**

OLS standard errors are 0.0963 on the intercept and 0.0966 on the slope.

Two entries need explanation. The OLS and normal MLE rows are identical in α and β because they are the same estimator: the only term in the normal log likelihood that involves β is $-\sum \epsilon_i^2/2\sigma^2$, so maximizing it over β *is* minimizing the sum of squares. They differ only in the variance estimate, and by exactly the expected factor: $1.3514 \times \sqrt{198/200} = 1.3446$. The OLS row divides by $n - p$ and is unbiased, the MLE row divides by n and is not.

The t row reports a **scale**, not a standard deviation. At $\nu = 3.6322$ the implied standard deviation is $\sigma\sqrt{\nu/(\nu-2)} = 0.9355 \times 1.4919 = 1.3956$, slightly *above* the normal's 1.3446. This distinction matters in part (e) and is the single easiest thing to get wrong in this problem.

AICc selects the t error model by a margin of 26.82. The t pays 2 extra AIC points for its fourth parameter and recovers 28.9 in likelihood.

3.3 Reconcile

(c) Compare the three estimates of β.

Table 5: The three slope estimates

Model	β	Difference from OLS	In OLS standard errors
OLS	2.9908	—	—
MLE, normal	2.9908	0.0000	0.000
MLE, t	3.0192	+0.0284	0.294

The prediction was right, and if anything understated. OLS and the normal MLE are numerically identical. The t estimate moves the slope by 0.0284, which is **0.29 of one OLS standard error** — well inside the noise of the estimate itself. A violation severe enough to be rejected at $p = 10^{-24}$ moved the parameter of interest by less than a third of its own standard error.

(d) What is the rejected model getting wrong, if not the slope?

The shape of the error distribution, and therefore everything downstream of it except the conditional mean.

The two models agree about where the line is. They disagree about how the observations are scattered around it, and that disagreement is what the likelihood is scoring. A ΔAICc of 26.82 corresponds to a log likelihood difference of 14.45, so the t assigns the observed residuals about $e^{14.45} \approx 1.9 \times 10^6$ times more joint density than the normal does. The data are nearly two million times more probable under the t — and the slope barely moved.

Concretely, what the normal model gets wrong is every quantity that depends on the error distribution beyond its first two moments:

- prediction and confidence intervals at any level in the tail;
- the probability of an exceedance of a given size, which is what a risk limit is;
- Value at Risk and Expected Shortfall computed from the fitted residual;
- the significance of the coefficients, since the t statistics assume normality.

What it gets right is the conditional mean. If the only thing wanted from this regression is a point forecast of y given x, the normal model is fine. If anything at all is wanted about the distribution around that forecast, it is not.

(e) Compare the 95% and 99.5% quantiles of the two fitted errors.

Table 6: Quantiles of the two fitted error distributions

Level	Fitted normal	Fitted t	Wider
95.0%	2.2117	2.0538	normal
97.5%	2.6354	2.7045	t
99.0%	3.1281	3.7098	t
99.5%	3.4635	4.6201	t

The normal is wider at 95%, the t is wider at 99.5%, and the two curves cross at about the 97.03% level.

The reason is that both distributions are fitted to the same 200 residuals, so they are forced to agree about where the bulk of the data sits. The t buys its heavy tails by pulling in its body: its scale is 0.9355 against the normal's 1.3446, a body 30% narrower. Out to roughly two standard deviations that narrower body dominates and the t is the *tighter* distribution. Past the crossover the t's polynomial tail decay, $(1 + z^2/\nu)^{-(\nu+1)/2}$, overtakes the normal's $e^{-z^2/2}$, and at $\nu = 3.63$ it does so quickly: by 99.5% the t is 33% wider.

This is the same trade Week 01 describes when it says the t has $\nu - 1$ finite moments and converges to the normal only as $\nu \to \infty$. A distribution fitted to data cannot be fatter everywhere; it can only redistribute mass, and the t redistributes it from the shoulders to the tails.

Which would I want for a capital buffer? The t, without hesitation.

A capital buffer is set at a far quantile — the Basel framework uses 99% VaR or 97.5% Expected Shortfall — and every level in Table 6 at or beyond 97.5% has the t wider. At 99.5% the normal would set a buffer of 3.4635 where the t calls for 4.6201: the normal buffer is 75% of the required size, a 25% shortfall on the number whose entire purpose is to survive the tail.

The 95% comparison is a trap worth naming explicitly. Read on its own it says the normal is the more conservative model, and that reading is exactly backwards for the use case. The model selection statistic and the tail behaviour agree with each other; only the 95% quantile, taken in isolation, points the other way, and 95% is not where a capital buffer lives.

4 Pearson Against Spearman

Code: `problem3.py`.

4.1 Predict

The pairwise plot is the whole prediction. Before computing any correlation, the marginal shapes of the four series are:

Table 7: Marginal shape of each series

Series	Mean	SD	Skewness	Excess kurtosis	Min	Max
`x1`	−0.0519	1.0108	0.079	0.039	−2.959	3.228
`x2`	−0.0804	3.8945	0.907	**21.992**	**−25.834**	**33.638**
`x3`	−0.0000	0.9845	−0.118	0.330	−3.293	3.607
`x4`	0.0757	0.9668	0.247	−0.035	−2.388	2.959

(a) Which pair will Pearson and Spearman disagree on most? Which will they agree on?

`x2` is unlike the other three. Its excess kurtosis is 22 against roughly 0 for the others, and its range runs to ± 30 where the others stop near ± 3. `x1`, `x3` and `x4` are all close to normal in shape.

Pearson is computed on values and is therefore sensitive to how far out the extreme observations sit. Spearman is computed on ranks and is not: an observation at $z = 8$ and an observation at $z = 3$ have adjacent ranks and contribute identically. So:

- **I expect the largest disagreement on a pair involving `x2`, and on `x1`–`x2` in particular.** In the scatter, `x1` and `x2` clearly move together, but the cloud is stretched vertically into a shape no straight line fits well. Pearson should come in materially below Spearman.
- **I expect agreement on every pair that does not involve `x2`.** `x1`–`x3`, `x1`–`x4` and `x3`–`x4` all pair two near normal margins, and with both margins well behaved the linear and the monotone summaries measure nearly the same thing.
- **`x4` looks independent of everything**, so both coefficients should be near zero for its three pairs and will trivially agree.

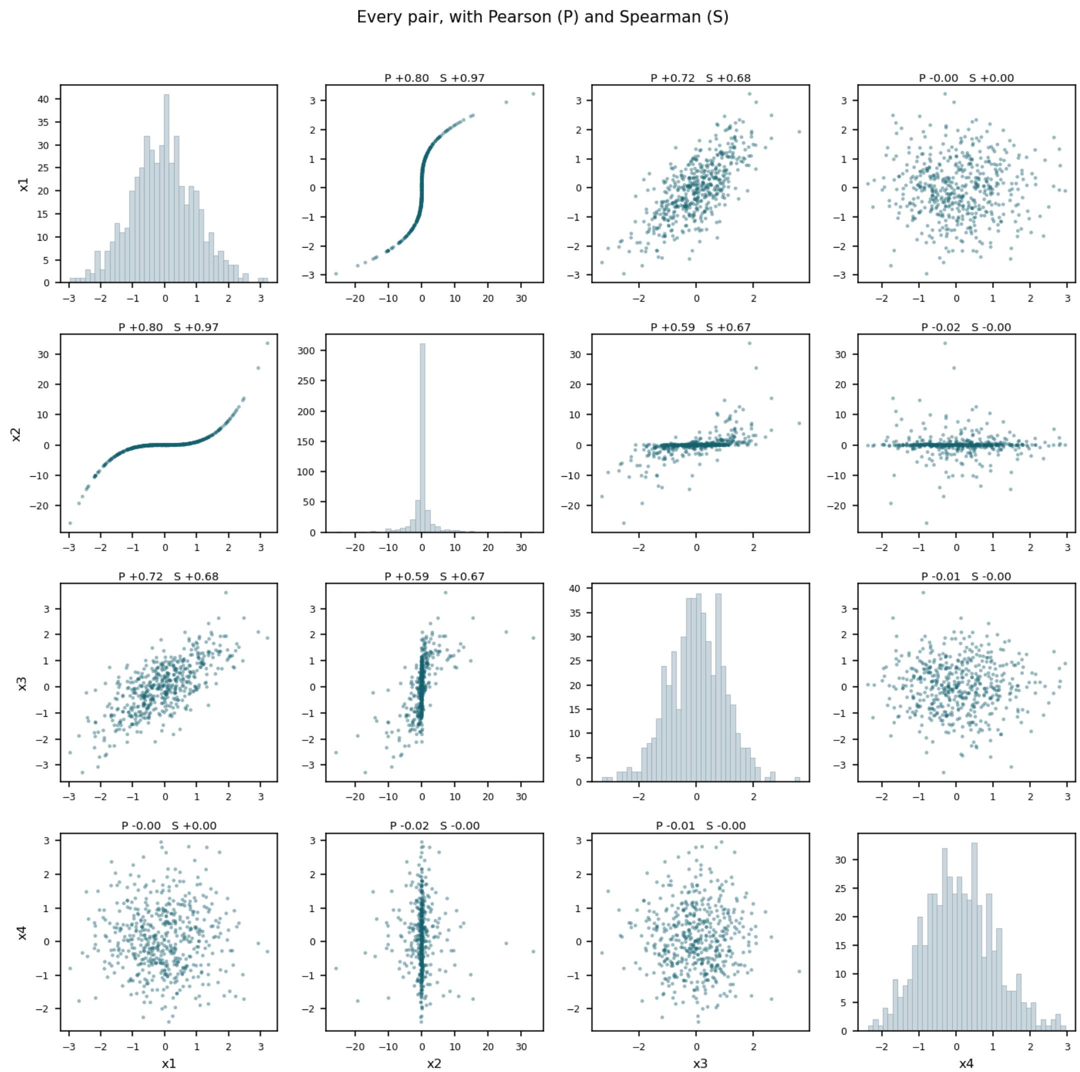

Figure 3: Every pair, with the Pearson (P) and Spearman (S) coefficient on each panel and the marginal histogram on the diagonal. The `x2` row and column are visibly different in scale from the rest.

4.2 Fit and reconcile

(b) Both correlation matrices, and the pair with the largest gap.

Pearson (lower triangle) and Spearman:

Table 8: Every pair, sorted by the size of the gap

Pair	Pearson	Spearman	|gap|
`x1–x2`	+0.7991	+0.9719	**0.1728**
`x2–x3`	+0.5876	+0.6674	0.0798
`x1–x3`	+0.7195	+0.6839	0.0356
`x2–x4`	−0.0209	−0.0011	0.0198
`x3–x4`	−0.0086	−0.0024	0.0062
`x1–x4`	−0.0047	+0.0002	0.0049

The largest gap is `x1–x2`: Pearson 0.7991 against Spearman 0.9719, a gap of 0.1728. The prediction holds in full. The two largest gaps are the two pairs involving `x2`, the three pairs involving `x4` are all near zero in both measures and agree to within 0.02, and `x1–x3` — two well behaved margins — agree to 0.036.

(c) What mechanism produces the gap?

Week 02 gives two distinct mechanisms that can open a Pearson–Spearman gap, and they call for different answers to the rest of this question, so the first job is to determine which one is operating. Each has a diagnostic.

Mechanism A, a handful of outliers. If a few extreme points are dragging Pearson, removing them should close the gap.

Table 9: Trimming the extremes does not close the gap

Trimming rule	Points dropped	Pearson	Spearman	Gap
none	0	0.7991	0.9719	0.1728
$\lvert z\rvert < 4$	5	0.8404	0.9711	0.1307
$\lvert z\rvert < 3$	12	0.8507	0.9698	0.1191
$\lvert z\rvert < 2.5$	21	0.8570	0.9681	0.1111

Trimming helps and then stops helping. Dropping the five most extreme observations moves Pearson by 0.04; dropping 21 — 4% of the sample, which is already more surgery than one would defend — moves it by 0.06 and leaves a gap of 0.111, still the largest in Table 8. **This is not an outlier story.**

Mechanism B, a monotone but non-linear relationship. If the dependence is monotone and the deficit in Pearson is the *shape of the margins*, then replacing both margins with normal scores — a rank preserving transform, so the dependence structure is untouched — should restore a high linear correlation.

Pearson on normal scores: 0.9845**.** Against a raw Pearson of 0.7991 and a Spearman of 0.9719. That is decisive.

The mechanism is therefore: **`x1` and `x2` have almost identical rank orderings, and the entire Pearson deficit is the marginal shape of `x2`.** Fitting a t to each margin makes the asymmetry explicit — `x1` returns $\nu = 177.9$, which is a normal for all practical purposes, while `x2` returns

$\nu = 0.485$, heavier tailed than a Cauchy. The right panel of Figure 4 shows the pair before and after the rank transform: an unusable cloud with a few points thrown to the corners becomes a tight, straight, near deterministic band.

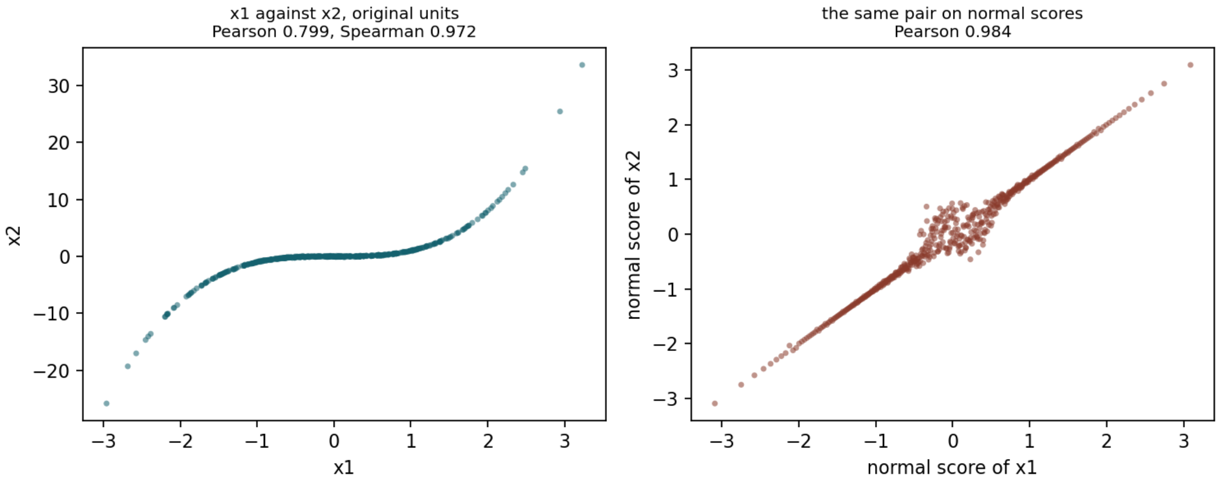

Figure 4: The widest pair before and after replacing both margins with normal scores. The rank transform preserves the dependence exactly and changes only the marginal shape, and the Pearson coefficient moves from 0.799 to 0.985.

Which coefficient is the honest description of this pair?

Spearman. The question anyone asks of a pair of series is whether they move together, and the honest answer here is that they move together almost without exception: when `x1` is high in its own distribution, `x2` is high in its own distribution, and the rank agreement is 0.97. Pearson's 0.80 understates that badly, and it understates it for a reason that has nothing to do with the relationship — it is an artifact of `x2` being measured on a scale with an enormously heavy tail.

What question is Pearson answering? "How well does a straight line, in the original units, describe this pair?" And 0.7991 is a *correct* answer to that question. A straight line genuinely does not describe these two series well, because `x2`'s extremes are so far out that no line can pass near both them and the body. Pearson is not wrong; it is answering a question about linear fit in raw units, and that question was not the one being asked.

One qualification, because it is the case where Pearson would be the number to use. If the purpose is to build a covariance matrix and compute a portfolio variance in the original units, the linear coefficient is the one that enters the arithmetic, and Spearman cannot be substituted for it. But with `x2`'s excess kurtosis at 22 the variance is a poor summary of `x2` in the first place. This is exactly the situation Week 05 resolves with a copula: use the rank correlation to describe the dependence, fit each margin its own distribution, and stop asking a single Pearson coefficient to carry both jobs.

5 Conditional Distributions

Code: `problem4.py`.

5.1 Predict

The sample covariance matrix of `problem4.csv`, $n = 1000$, rows and columns ordered `x1`, `x2`:

$$\hat{\Sigma} = \begin{pmatrix} 1.099875 & 1.696902 \\ 1.696902 & 4.104749 \end{pmatrix}, \qquad \hat{\rho} = 0.798623$$

Block mapping. The Week 02 result partitions X into x_1 and x_2, conditions on $x_2 = a$, and returns the distribution of x_1. This problem conditions on `x1` and asks about `x2`, which is the other way round, so the blocks map as:

Table 10: Which series is which block

Week 02 block	This problem	Value
Σ_{11} (variable of interest)	$\mathrm{var}(\mathtt{x2})$	4.104749
Σ_{22} (conditioned on)	$\mathrm{var}(\mathtt{x1})$	1.099875
Σ_{12} (cross block)	$\mathrm{cov}(\mathtt{x2}, \mathtt{x1})$	1.696902

(a) The conditional variance, and the factor by which uncertainty falls.

$$\bar{\Sigma} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = \mathrm{var}(\mathtt{x2}) - \frac{\mathrm{cov}(\mathtt{x1}, \mathtt{x2})^2}{\mathrm{var}(\mathtt{x1})} = 4.104749 - \frac{1.696902^2}{1.099875} = 1.486746$$

As a formula in the same blocks, the factor is

$$\frac{\bar{\Sigma}}{\Sigma_{11}} = 1 - \frac{\Sigma_{12}^2}{\Sigma_{11}\Sigma_{22}} = 1 - \rho^2$$

and for this data $1 - 0.798623^2 = \mathbf{0.362201}$.

So learning `x1` removes **63.8%** of the variance of `x2`. In standard deviation terms the conditional σ is $\sqrt{1.486746} = 1.219322$ against an unconditional 2.026018, a reduction to $\sqrt{0.362201} = \mathbf{60.18}\%$ of what it was.

(b) Does that factor depend on the value of `x1` observed?

No. The term that settles it is the *absence* of a term: the conditioning value a appears in

$$\bar{\mu} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(a - \mu_2)$$

and appears nowhere in

$$\bar{\Sigma} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}.$$

Under the multivariate normal the conditional variance is built out of the covariance structure alone. Observing `x1` at its mean and observing it four standard deviations away reduce the uncertainty about `x2` by exactly the same amount. This is answered before looking at the data because it is a structural property of the family, not a property of any sample.

It is also the property that part (f) turns out to depend on.

5.2 Fit and reconcile

(c) The conditional mean.

$$\bar{\mu} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(a - \mu_2) = \overline{\texttt{x2}} + \frac{\mathrm{cov}(\texttt{x1}, \texttt{x2})}{\mathrm{var}(\texttt{x1})}\,(\texttt{x1} - \overline{\texttt{x1}})$$

The coefficient on `x1` is $\Sigma_{12}\Sigma_{22}^{-1} = 1.696902/1.099875 = \mathbf{1.542813}$, with an implied intercept of 0.071073.

In regression terms this coefficient is the OLS slope of `x2` on `x1`. Not an analogue of it — the same number. Week 02 derives $\hat{\beta}_i = \mathrm{cov}(x_i, y)/\sigma^2_{x(i)}$ from a minimization problem and $\Sigma_{12}\Sigma_{22}^{-1}$ from conditioning a normal vector, and they are one object arriving through two doors. The script fits the OLS line independently as a check: it returns a slope of 1.542813, agreeing to 4.4×10^{-16}.

The 95% band is $\bar{\mu} \pm 1.96\sqrt{\bar{\Sigma}} = \bar{\mu} \pm 2.389871$. **Its half width is a constant**, precisely because of part (b): $\bar{\Sigma}$ does not contain a, so the band drawn in Figure 5 has parallel edges.

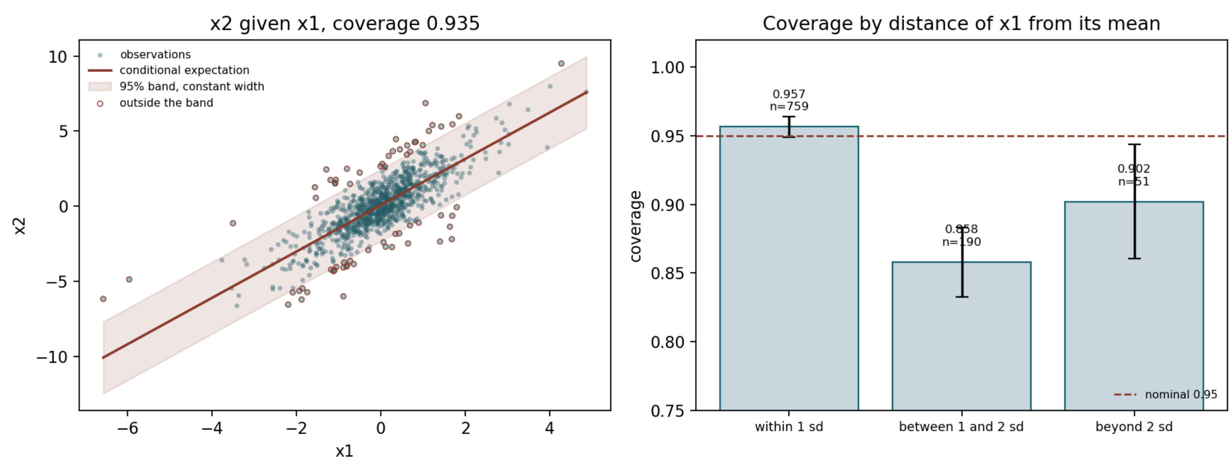

Figure 5: Left: the conditional expectation of `x2` given `x1` with its constant width 95% band over the scatter; the 65 observations outside the band are circled. Right: coverage within each bucket, with binomial error bars, against the nominal 0.95.

(d) What fraction of observations fall inside the band?

935 of 1000, or 0.9350**.** The nominal level is 0.95. The binomial standard error at the nominal level is 0.0069, so the shortfall is 2.18 standard errors — noticeable, but on its own not dramatic. The bucketed version below is where the real problem shows.

(e) Coverage by how far `x1` sits from its mean.

Table 11: Coverage within each bucket

Bucket	n	Coverage	SE	Residual SD
within 1 sd	759	0.9565	0.0074	1.0758
between 1 and 2 sd	190	0.8579	0.0253	1.5804
beyond 2 sd	51	0.9020	0.0416	1.5957

The band assumes a single conditional standard deviation of 1.2193 everywhere. The last column shows what the residuals actually do.

(f) What does the non-flat coverage tell us, and which assumption failed?

The pattern. Coverage is *above* nominal near the centre of `x1` and *below* it away from the centre, and the residual standard deviation grows monotonically across the buckets — 1.076, 1.580, 1.596 — against the pooled 1.219 the band assumes. The conditional spread of `x2` is not constant: **it grows with how far `x1` sits from its mean.** In the middle the band is too wide, so it over-covers; in the tails it is too narrow, so it under-covers.

The assumption that failed is joint normality, and specifically the consequence of it established in part (b): that the conditional variance does not depend on the conditioning value. That is the sole reason the band was drawn with constant width. Remove it and the constant width band has no justification, which is what Table 11 is measuring.

Supporting evidence that the joint normal is not the right family: the residual has excess kurtosis +1.5130 and Jarque–Bera $p = 1.4 \times 10^{-21}$, and neither margin is normal either (`x1` has excess kurtosis +3.872, `x2` +1.521). A joint normal has normal margins; these do not.

A family that does reproduce the pattern. The natural candidate is a bivariate t, and fitting one turns "something like a t" into a number that can be checked. Maximum likelihood gives $\nu = 4.490$ and a scale matrix whose implied covariance $\frac{\nu}{\nu-2}\Sigma$ is

$$\begin{pmatrix} 1.094994 & 1.749290 \\ 1.749290 & 4.404128 \end{pmatrix} \quad \text{against the sample} \quad \begin{pmatrix} 1.099875 & 1.696902 \\ 1.696902 & 4.104749 \end{pmatrix}$$

with an implied correlation of 0.7966 against the sample 0.7986. Under a bivariate t the conditional standard deviation of `x2` given `x1` carries a multiplier $\sqrt{(\nu + d_1)/(\nu + 1)}$, where d_1 is the squared standardized distance of the conditioning value from its mean — **a conditional variance that does depend on the observed `x1`**, which is exactly the property the normal forbids and the data displays. Evaluated at the fitted ν:

Table 12: Bivariate t prediction against the observed bucket residuals

Bucket	Typical $\lvert z\rvert$	Predicted SD	Observed SD
within 1 sd	0.5	1.1330	1.0758
between 1 and 2 sd	1.4	1.3216	1.5804
beyond 2 sd	2.5	1.7055	1.5957

The direction and rough magnitude match; the middle bucket comes in above the prediction and the outer bucket below it. I would not claim more than "the right shape" from this, for the reason in the caveat below.

Which parts of (c) survive, and which do not. The assignment asks to be careful here, and the split is clean:

- **The conditional mean survives, coefficient and all.** Two independent reasons. First, the bivariate t is an *elliptical* distribution, and every elliptical distribution has a conditional mean that is exactly linear in the conditioning variable with the same coefficient $\Sigma_{12}\Sigma_{22}^{-1}$. Second, and more generally, that coefficient is the best linear predictor of `x2` given `x1` for *any* joint distribution with finite second moments — it is a statement about second moments and requires no distributional assumption at all. The line drawn in the figure is the right line.
- **The constant width of the band does not survive.** It rests entirely on $\bar{\Sigma}$ being free of a, which is a property of the normal and not of the data.
- **The 95% claim does not survive either.** Even a band correctly widened with $|x_1|$ would need the conditional *distribution* to be normal for 1.96 to be the right multiplier, and the residual's excess kurtosis of +1.51 says it is not.

So what is left of part (c) is the regression line. What is lost is everything about the uncertainty around it.

One caveat on reading Table 11. Coverage is not monotone across the three buckets: the middle bucket (0.8579) sits below the outer one (0.9020). With only 51 observations beyond two standard deviations, the standard error there is 0.0416, so those two figures are about one standard error apart and the ordering between them should not be interpreted. The signal that is robust is the contrast between the inner bucket, which over-covers by more than its own standard error, and the two outer buckets, which under-cover.

6 Identifying an AR or MA Order

Code: `problem5.py`.

6.1 Predict

$n = 500$, mean 0.6709, standard deviation 1.1211. The significance band is $\pm 1.96/\sqrt{500} = \pm 0.0877$.

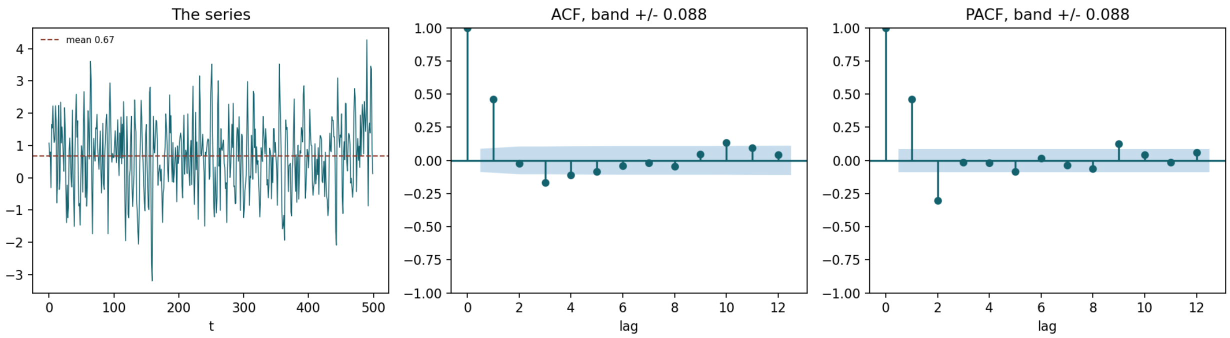

Figure 6: The series with its ACF and PACF. The shaded region on the two right panels is the $\pm 1.96/\sqrt{n}$ band. The PACF returns inside it at lag 3 and stays; the ACF does not.

Table 13: ACF and PACF, first eight lags

Lag	ACF	Outside band	PACF	Outside band
1	+0.4616	yes	+0.4616	yes
2	−0.0254		−0.3031	**yes**
3	−0.1657	yes	−0.0147	
4	−0.1104	yes	−0.0169	
5	−0.0841		−0.0843	
6	−0.0405		+0.0159	
7	−0.0193		−0.0381	
8	−0.0469		−0.0632	

(a) Which cuts off, which decays, and what order?

The PACF cuts off after lag 2. The ACF decays. That is an **AR(2)**, and I commit to it before fitting.

The PACF is significant at lags 1 and 2 and then falls inside the band at lag 3 and stays there through lag 8 — −0.0147, −0.0169, −0.0843, +0.0159, −0.0381, −0.0632, none of them outside ±0.0877. That is a cut-off.

The ACF does not cut off. It starts at +0.4616, changes sign, and continues to produce non-zero values at lags 3 and 4 that are still outside the band before shrinking. A sign changing, shrinking sequence is a decay, not a cut-off. The decay is *oscillating* rather than the geometric $\rho(k) = \beta^k$ of an AR(1); that is a hint the AR polynomial has complex roots, which the fit confirms below.

(b) The rule, and the band.

The rule is the Week 02 table:

Table 14: The identification rule

Process	ACF	PACF
MA(q)	cuts off after lag q	decays
AR(p)	decays	cuts off after lag p
ARMA(p, q)	decays	decays

A function "cuts off after lag k" when its sample values fall inside the $\pm 1.96/\sqrt{n}$ band from lag $k+1$ onward and stay there. Here $n = 500$, so the band I judged against is $\pm 1.96/\sqrt{500} = \pm\mathbf{0.0877}$. The PACF's last significant lag is 2 at -0.3031, three and a half times the band, and lag 3 at -0.0147 is a sixth of it. The transition is not marginal.

An honest caveat. Extending the table to twelve lags, the ACF is outside the band again at lags 10 and 11 (+0.1321, +0.0925) and the PACF at lag 9 (+0.1264). With twelve lags tested against a 5% band, some excursions are expected by chance, and these are isolated, at long lags, and small. I read them as noise and identify the model from the low order structure, but the picture is not perfectly clean.

6.2 Fit and reconcile

(c) AICc for six models.

Using $k = p + d$: an AR(p) has p autoregressive coefficients plus an intercept, so $p_{\text{reg}} = p + 1$, plus $d = 1$ for the fitted variance, giving $k = p + 2$. Likewise $k = q + 2$ for MA(q).

Table 15: AICc for all six models

Model	$\ln L$	k	AIC	AICc	ΔAICc
AR(1)	−706.2693	3	1418.5386	1418.5870	46.1486
AR(2)	−682.1788	4	1372.3575	**1372.4383**	**0.0000**
AR(3)	−682.1109	5	1374.2218	1374.3432	1.9049
MA(1)	−691.5179	3	1389.0359	1389.0843	16.6460
MA(2)	−686.5754	4	1381.1508	1381.2316	8.7933
MA(3)	−681.9975	5	1373.9951	1374.1165	1.6782

(d) Which model does AICc select? Did it match the prediction?

AICc selects AR(2), and it matches the prediction exactly.

The fitted coefficients are $\phi_1 = +0.6012$ (se 0.0424), $\phi_2 = -0.3026$ (se 0.0431), with an intercept of 0.6706 and $\sigma^2 = 0.8958$. Both autoregressive coefficients are many standard errors from zero.

The second coefficient also explains the shape noted in part (a). The characteristic equation $z^2 - 0.6012z + 0.3026 = 0$ has discriminant $\phi_1^2 + 4\phi_2 = -0.8491 < 0$, so the roots are the complex pair $0.3006 \pm 0.4607i$ with modulus 0.5501. A complex root pair produces a *damped oscillation* in the autocorrelation function rather than a monotone decay, which is why the ACF changes sign at lag 2 instead of shrinking steadily. The modulus is inside the unit circle, so the process is stationary.

One honest observation about Table 15: **MA(3) is close behind**, at ΔAICc $= 1.68$. That is not a large margin. An AR(2) with complex roots and an MA(3) can generate similar low order autocorrelation patterns, so it is unsurprising that the data does not separate them sharply. The identification argument from the PACF is what breaks the tie, and it points at AR(2) unambiguously; AICc agrees, but it agrees by less than the raw ordering suggests.

(e) AR(2) against AR(3): what is AICc trading off, and why would R^2 not make the same choice?

The third autoregressive coefficient is -0.01654 with a standard error of 0.04380, a t statistic of -0.378. It is indistinguishable from zero.

What AICc trades off. Adding it raises the log likelihood from -682.1788 to -682.1109, a gain of 0.0679, which enters AICc as $2 \times 0.0679 = 0.136$. Against that, moving from $k = 4$ to $k = 5$ costs 2 in the $2k$ term plus an increase in the correction $\frac{2k^2+2k}{n-k-1}$. The charge is roughly 2.04 and the gain is 0.14, so AICc rises by 1.9049 and the larger model is rejected. That is the whole trade: **a fixed price per parameter against whatever fit the parameter buys, and this parameter bought almost nothing.**

Why R^2 would not make the same choice. R^2 rose, from 0.285730 to 0.285925, a gain of 1.95×10^{-4}. It had to. Adding a regressor can never increase the residual sum of squares, because the larger model contains the smaller one as the special case where the extra coefficient is zero, so R^2 is non-decreasing in the number of parameters by construction. It carries no penalty term at all. Applied as a selection rule it would choose AR(3) over AR(2), and would go on choosing the largest model available no matter how many lags were offered — the same failure Week 02 describes when it notes that adding a column of random numbers raises the R^2 of any regression.

That is the difference between a measure of fit and a criterion for selection. R^2 answers "how much of the variance did this model account for in this sample", and 0.2859 is an honest answer to that. AICc answers "is this parameter worth having", and for the third autoregressive coefficient the answer is no.